CR-BIS Data Acquisition User Guide

Document Version: 2.0

Contents

1. Introduction

This guide describes the use of the ***CR-BIS (Continuous Recording Beam-Image Shift)*** data acquisition script for **SerialEM**. Users can customize parameters as needed and perform multiple acquisition modes, including conventional BIS and CR-BIS. The following sections focus on the operation of the CR-BIS mode and its required pre-calibration steps.

Part 1: Pre-calibration

1.1 Measuring Camera Delay Curves

Different cryo-EM camera models exhibit distinct delay characteristics. It is recommended to measure the delay curve in advance to quantify the theoretical acceleration achievable with CR-BIS.

Use the script **Report_Clock_script.txt**. Set a series of exposure times and perform five repeated measurements for each setting. Plot the relationship between the set exposure time and actual recorded time to obtain the intercept and slope parameters.

```
start 0.3  s
 3 frames were saved to X:\userdata\20250520_yq_test\K2_0006.tif
2.53 seconds elapsed time
 3 frames were saved to X:\userdata\20250520_yq_test\K2_0007.tif
2.53 seconds elapsed time
 3 frames were saved to X:\userdata\20250520_yq_test\K2_0008.tif
2.50 seconds elapsed time
 3 frames were saved to X:\userdata\20250520_yq_test\K2_0009.tif
2.63 seconds elapsed time
 3 frames were saved to X:\userdata\20250520_yq_test\K2_0010.tif
2.56 seconds elapsed time
start 0.6  s
 6 frames were saved to X:\userdata\20250520_yq_test\K2_0011.tif
2.84 seconds elapsed time
 6 frames were saved to X:\userdata\20250520_yq_test\K2_0012.tif
2.83 seconds elapsed time
 6 frames were saved to X:\userdata\20250520_yq_test\K2_0013.tif
2.81 seconds elapsed time
 6 frames were saved to X:\userdata\20250520_yq_test\K2_0014.tif
2.83 seconds elapsed time
 6 frames were saved to X:\userdata\20250520_yq_test\K2_0015.tif
2.83 seconds elapsed time
start 1  s
 10 frames were saved to X:\userdata\20250520_yq_test\K2_0016.tif
3.27 seconds elapsed time
 10 frames were saved to X:\userdata\20250520_yq_test\K2_0017.tif
3.25 seconds elapsed time
 10 frames were saved to X:\userdata\20250520_yq_test\K2_0018.tif
3.34 seconds elapsed time
 10 frames were saved to X:\userdata\20250520_yq_test\K2_0019.tif
3.23 seconds elapsed time
 10 frames were saved to X:\userdata\20250520_yq_test\K2_0020.tif
3.27 seconds elapsed time
```

SerialEM log output example

1.2 Determining the Formula for CR-BIS Total Exposure Time

In CR-BIS mode, the total set exposure time is defined as:

$$T_{total\ set} = (T_{set\ per\ point} + T_{blank}) * point\ number + T_{buffer} \quad (1)$$

Examples:

K3 camera: $T_{total\ set} = (T_{set\ per\ point} + 0.22\ s) * point\ number + 0.6\ s)$

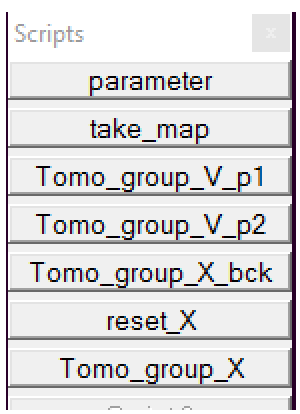
Falcon 4i camera: $T_{total\ set} = (T_{set\ per\ point} + 0.26\ s) * point\ number + 1.2\ s)$

A CR-BIS acquisition collects multiple target points in a single continuous exposure. By analyzing frame rate and blank-frame data, one can estimate the beam blank/shift time and buffer duration.

Part 2: CR-BIS Data Acquisition Workflow

2.1 Overview of CR-BIS Scripts and Toolbar

The CR-BIS package includes several sub-scripts accessible through the SerialEM toolbar:



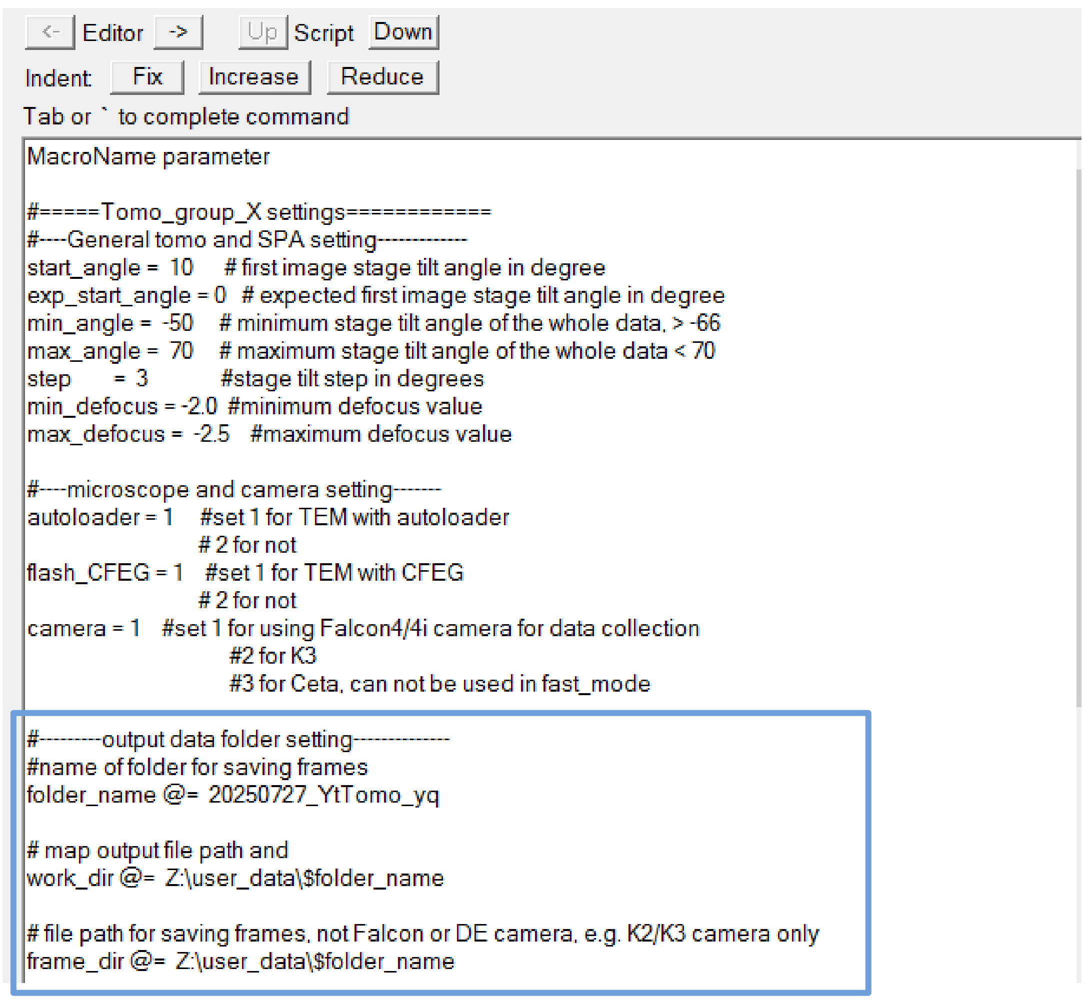
- **parameter** — Core parameter configuration script; must be executed before running other scripts.
- take_map — Captures low-magnification maps for target selection.
- Tomo_group_V_p1/p2 — Performs precise alignment of target points on the low-magnification map (critical for cryo-ET).

- reset_X — Records key microscope parameters after setup and restores them before acquisition to ensure consistency.
- Tomo_group_X — Main data acquisition script for production runs.

2.2 Parameter Configuration

The parameters required for CR-BIS operation are shown in the figure below. Each colored section corresponds to a specific parameter group in the parameter script.

Script 1: parameter



Blue Box — Output Paths

Specify the output directory for data files.

Ensure that the map files and data output files are stored in separate directories to avoid overwriting or file management conflicts.

```
#--------data collection mode setting--------------
data_mode =  1  #set 1 for dose sym tomography
                  #2 for one or two direction tomography
                  # 3 Single particle tilt to fix angle for data collection
                  #4 Single particle tilt pair
fast_mode = 1       # 1 for using fast mode, SET FAST MODE as well
                  # 2 for not
accurate_position = 1  #1 for aligning each point to reference in V, 2 for not
use_focus_forZ = 2  #set 1 for focus every point to measure Z
                  #2 for not
do_eucentric_fine = 1  #set 1 for do eucentric fine each group
                  #2 for not
align_low_angle_only = 1   #1 for only align at first 4 tilts
                  #2  for align at all tilts

skip_value_limit  = -100  # Value of buffer mean count to skip, lower than this value will skip collection
                  #use "ReportMeanCounts" to check value

backgroud_stage = 1  #1 for use file_checker for stage movement when imaging
                  #2 for not

move_stage_instead_IS  = 2  # 1 for only moving stage, 2 for using image shift besides stage
                  # 3 for only using image shift at high mag


#------multi shot per hole for SPA setting-----
in_hole_distance = 0   # distance away from the selected point in micrometer, only for SPA, not for tomography and tilt pair
in_hole_num = 1         # num of image taken around the  selected point, only for SPA, not for tomography and tilt pair
in_hole_start_angle = 0       # angle for the first point away from selected point, the only for SPA, not for tomography and tilt pair

#-------FAST MODE setting-------
R_delay = 0.9   # exposure time for Record mode in FAST mode
blank_delay = 0.72   # delay time in sec for the beginning of continous record mode
ObjVsISmatrix = { -0.000005  0.000049  -0.000011  -0.000050 }  #do Focus/Tune -> "Calibrate Coma vs Image Shift" and cope the output ObjVsISmatrix in log
```

Red Box — CR-BIS Core Parameters

To enable CR-BIS mode, set: fast_mode = 1

- **R_delay** — Exposure time for each target point under CR-BIS mode.
- **blank_delay** — A microscope-dependent parameter that compensates for camera model–specific delays.

  - For K3 cameras: when switching to Continuous Mode, a short delay occurs that may cause the first target exposure to be skipped. This value should therefore be included as <blank_delay>.

  - For Falcon4i cameras: there is no switching delay. However, the beam is briefly turned on at the focus position before the first target exposure, requiring an additional buffering time in the total exposure time formula.
- **ObjVsISmatrix** — Calibration matrix between objective lens deflection and image shift, used for coma and image-shift correction.
- For CR-BIS-Tomo mode, set **align_low_angle_only = 1**. This ensures the first five tilt angles use standard acquisition to obtain better alignment parameters.

Yellow Box — Fine Alignment Parameters

These parameters improve positional accuracy and are particularly important for cryo-ET.

• **accurate_position** — Optional for SPA acquisition; recommended for cryo-ET.

• **use_focus_forZ** — Measures focus at each target point at the starting tilt angle to obtain Z-height.

  - Ensure the focus position differs from the actual imaging region.
  - Recommended for grid-based cryo-ET; not suitable for FIB-lamellae.

• **do_eucentric_fine** — Performs fine eucentric height adjustment.

  - Recommended for non-lamella cryo-ET samples.
  - Not recommended for lamella-based cryo-ET due to potential occlusion at high tilt angles.

Purple Box — Background Stage Movement

The background_stage parameter allows pre-movement of the sample stage while CR-BIS exposures are running. This utilizes the idle camera buffering period after a long CR-BIS exposure to move the stage to the next tilt angle or focus group, saving acquisition time. To enable this feature:

1. Copy **file_checker.py** to the directory where the map files are stored.
2. Before starting acquisition, run:

python3 file_checker.py

This script monitors the exposure progress and automatically triggers stage movement once the exposure group finishes.

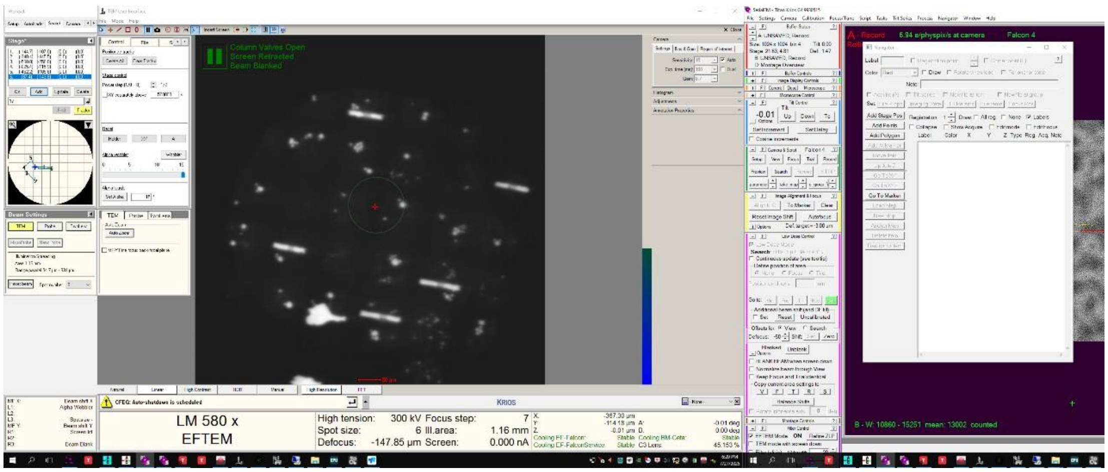
Initialization

After all parameters have been properly set, run the parameter script once. This will initialize the configuration and automatically create a data acquisition directory under the specified path for the current data collection session.

2.3 Low-Magnification Map Acquisition

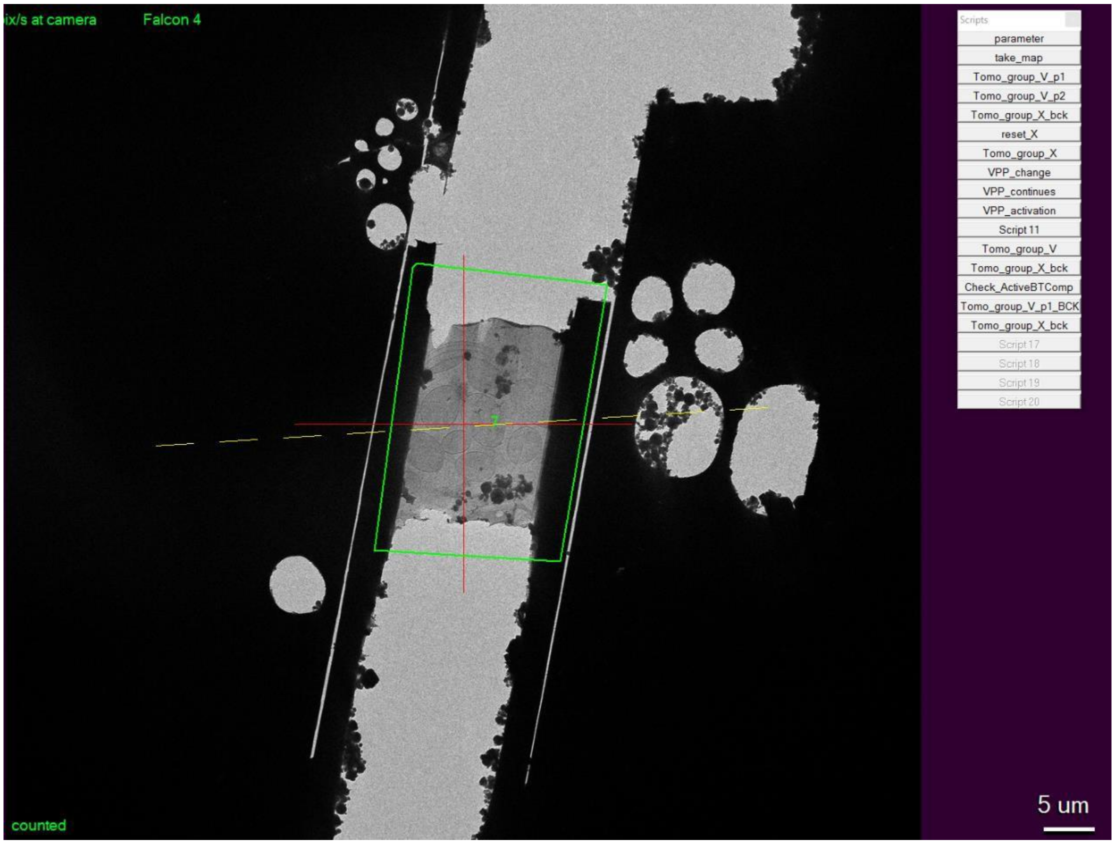
The procedure for acquiring low-magnification maps in **CR-BIS** is essentially identical to conventional cryo-EM data collection. The steps are as follows:

1. At low magnification, locate the target **mesh** or **lamella**.
2. Adjust the Z-height manually or automatically using **Tasks → Eucentric** to bring the specimen to the eucentric position.



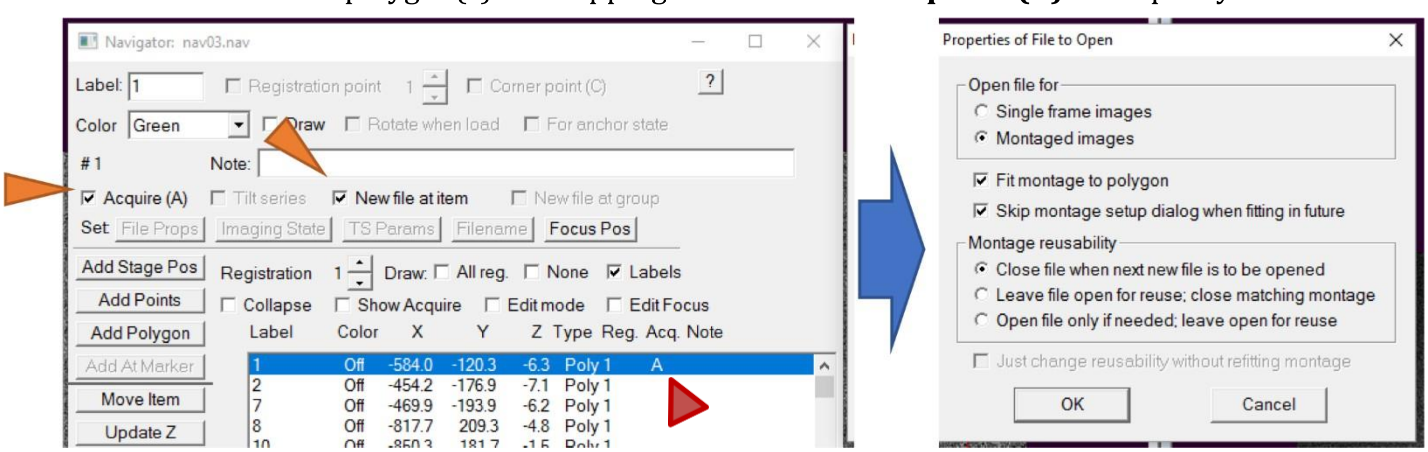
Low-magnification overview showing mesh or lamella locations. Green boxes indicate areas selected with Navigator → Add Polygon for subsequent high-magnification data collection.

3. In the **Navigator**, outline the desired acquisition regions using **Add Polygon**. Each polygon defines a region (e.g., one lamella or grid square) for subsequent targeting.

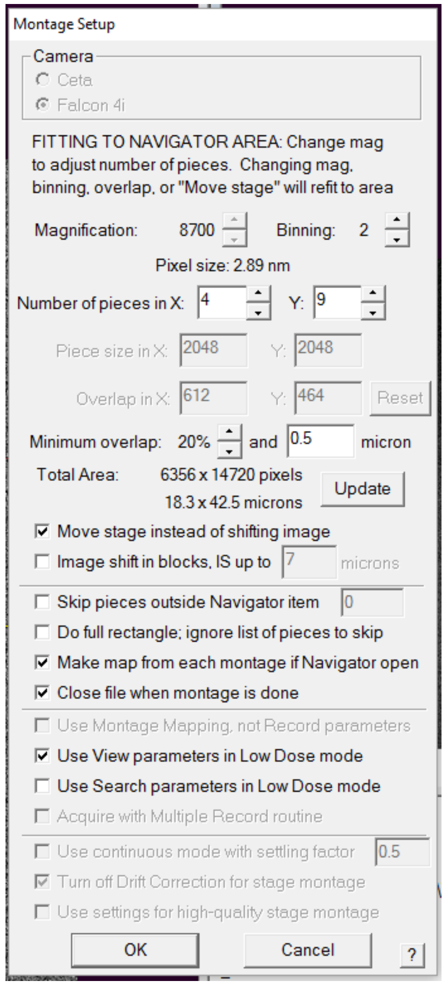


A representative lamella imaged at low magnification. The green polygon outlines the acquisition area; red or yellow annotations mark excluded or reference regions.

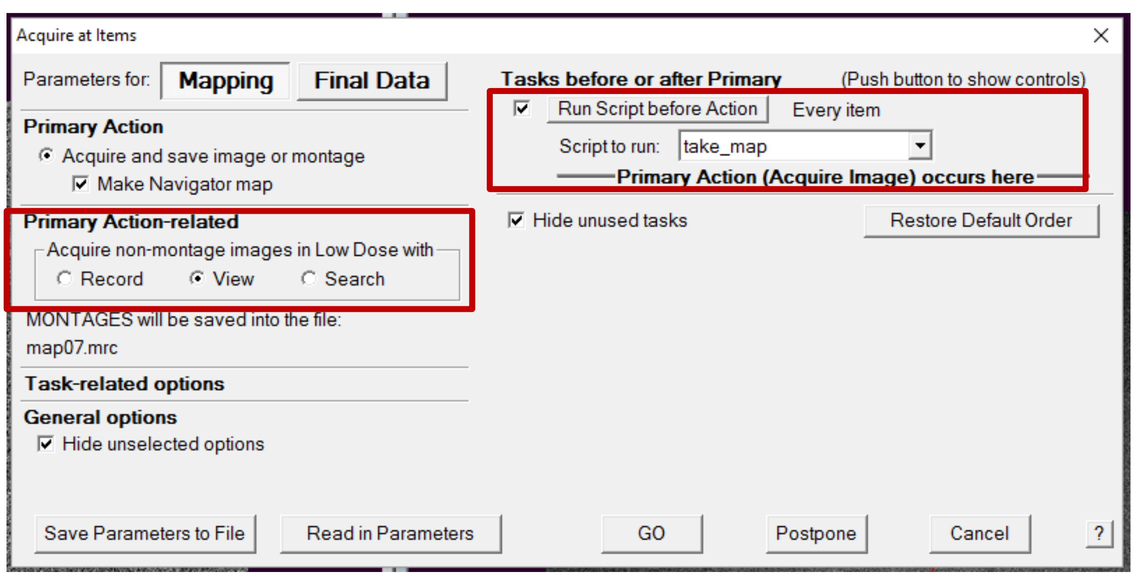
4. Switch to **View** mode, and set exposure parameters such as exposure time, binning, and gain reference.
5. Select the polygon(s) for mapping and check the **"Acquire" (A)** box. Specify an



Left: start of a new target group, showing the first (focus) point and subsequent acquisition points. Right: end of the same group after all points are defined.



6. Execute **Navigator → Acquire at Items** to record the low-magnification map.



2.4 Target Selection

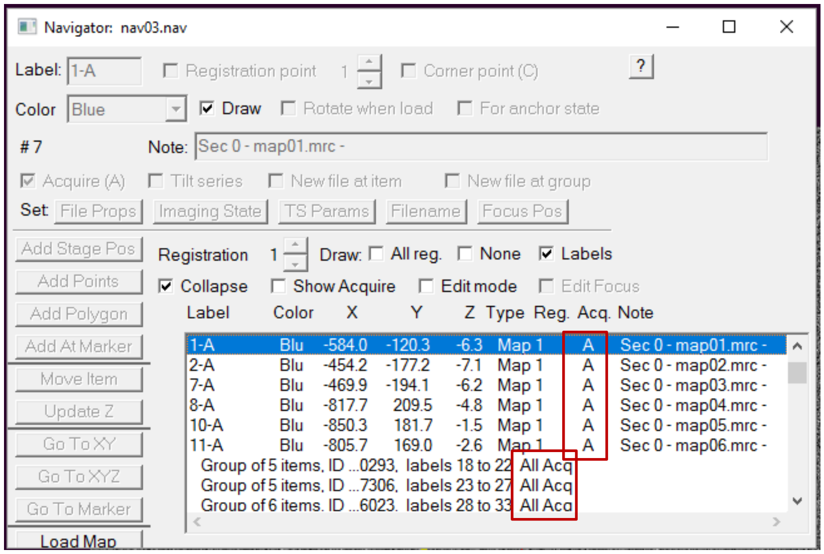
Once maps are acquired, define imaging targets in groups for automated **CR-BIS** collection.

1. Double-click the **map** in the Navigator to open it.
2. Select targets either manually or automatically, organizing them into **groups**. Each group should contain several nearby points.
3. The **first point of each group** serves as the *alignment/focus position* and is **not recorded** as data.
4. The remaining points in the group are acquisition targets. Recommended distances between the first and subsequent points: **Lamella (cryo-ET)**: < 6 µm; **Single-particle (SPA)**: up to 10 µm (maximum practical distance) Distances should respect the beam-shift and coma-correction range.
5. Continue defining new groups until all target regions are covered.

2.5 Production Data Acquisition

After all maps and targets are defined, proceed with **CR-BIS data acquisition**:

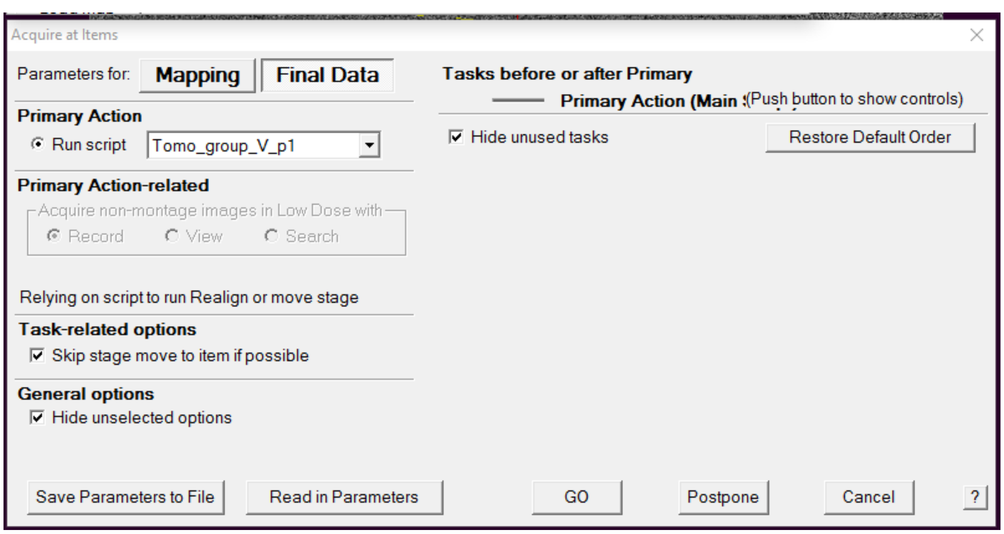
1. Verify that all maps and points are **checked with "A"** in the Navigator. If polygons also have "A" marks, **remove them** (only points should remain checked).



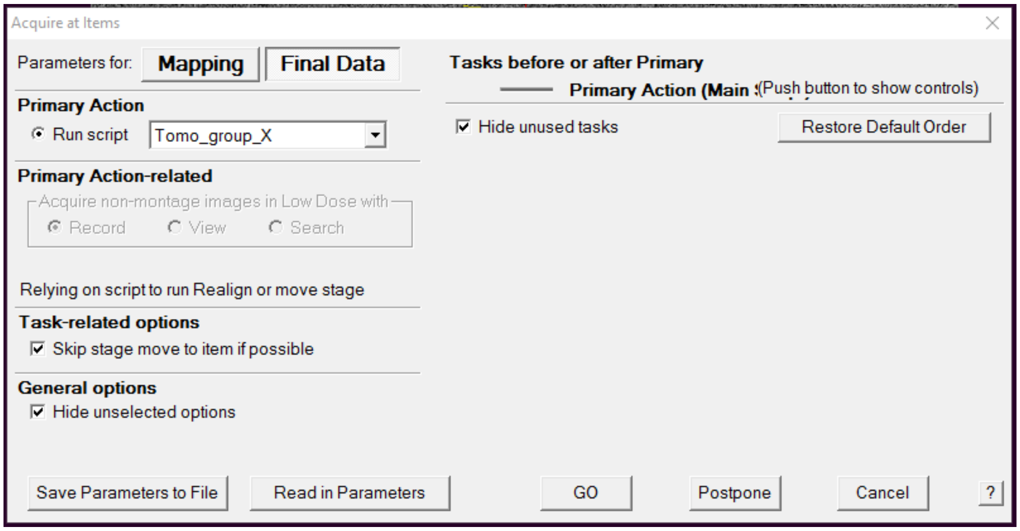
Ensure that each **Map item precedes its corresponding Points** in the Navigator list (maps must appear above their points).

2. Run the pre-alignment script via **Acquire at Items → FinalData(Tomo_group_V_p1)** to initialize alignment and metadata recording.

    - If acquisition is interrupted and the order of maps or points changes, **delete map_info.txt** before resuming to avoid indexing errors.



3. Start the main acquisition run using **Acquire at Items → FinalData(Tomo_group_X)**.



4. Monitor the acquisition log to confirm that data writing proceed normally.

Part 3: Preprocessing of CR-BIS Output Data

All preprocessing scripts require Python 3.

3.1 Processing TIF-format Data

Script: group_single_point_tif_MotionCorr2_parallel_v2_tiffile.py

Usage:

python3 group_single_point_tif_MotionCorr2_parallel_v2_tiffile.py [options]

input_directory

Generates separated single-point files.

Required: input_directory (path to raw CR-BIS data).

Threshold (default 0.1)

3.2 Processing EER-format Data

```
20260109_170347_EER_GainReference.gain
Falcon4_00000.eer
Falcon4_00001.eer
Falcon4_00002.eer
Falcon4_00003.eer
Falcon4_00004.eer
Falcon4_00005.eer
Falcon4_00006.eer
Falcon4_00007.eer
Falcon4_00008.eer
Falcon4_00009.eer
Falcon4_00010.eer
Falcon4_00011.eer
Falcon4_00012.eer
Falcon4_00013.eer
Falcon4_00014.eer
Falcon4_00015.eer
Falcon4_00016.eer
Falcon4_00017.eer
Falcon4_00016.eer
Falcon4_00017.eer
Falcon4_00018.eer
Falcon4_00019.eer
Falcon4_00020.eer
Falcon4_00021.eer
Falcon4_00021.eer_log
Falcon4_00022.eer
Falcon4_00022.eer_log
Falcon4_00023.eer
Falcon4_00023.eer_log
Falcon4_00024.eer
Falcon4_00024.eer_log
Falcon4_00025.eer
Falcon4_00025.eer_log
Falcon4_00026.eer
Falcon4_00026.eer_log
Falcon4_00027.eer
Falcon4_00027.eer_log
```

First 5 tilts BIS outputs

CR-BIS outputs

CR-BIS-Tomo Output Folder Example (Before Splitting)

```
cat Falcon4_00021.eer_log
```

```
5.99553 -0.284781 -1.791169 4 0.041321 6 34 R 29471278
5.99553 -1.637413 1.343705 5 0.020546 6 34 R 29471278
5.99553 2.867066 -0.467366 6 0.016017 6 34 R 29471278
5.99553 1.537317 2.590912 7 0.007401 6 34 R 29471278
```

The acquisition parameters for each target position within a CR-BIS output file are recorded in the corresponding ***.eer_log** file.

Step 1: Calculate blank frame gap before splitting EER files

Copy several CR-BIS output files into a test folder (increasing the number of files generally improves the accuracy of the distribution statistics), and run:

```
python3 calculate_sep_range_python34.py
```

```
 Falcon4_00021.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00021.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 60;
min gap between files is 45, max gap is 47;
min frame number in each file is 26, max is 68
 Falcon4_00022.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00022.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 60;
min gap between files is 47, max gap is 48;
min frame number in each file is 23, max is 69
 Falcon4_00023.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00023.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 60;
min gap between files is 46, max gap is 47;
min frame number in each file is 26, max is 67
 Falcon4_00024.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00024.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 61;
min gap between files is 45, max gap is 46;
min frame number in each file is 25, max is 68
```

blank frame number in this eer file

……

```
 Falcon4_00054.eer : processing...
/media/yq/Elements/20260113_fast_tomo/test/Falcon4_00054.eer: 826 frames separate into 4 file. First file starts at 401, last file ends at 63;
min gap between files is 47, max gap is 48;
min frame number in each file is 21, max is 68
first_range:(401, 401)
blank_range:(44, 52)
end_range:(53, 71)
```

The statistics of all files in the current directory

For each CR-BIS output EER file, the calculated blank frame gaps will be written into ***_json_dose_sep.**

```
cat Falcon4_00021_json_dose_sep
```


```
{
    "starts": [
        359,
        472,
        586,
        700
    ],
    "ends": [
        428,
        541,
        655,
        769
    ]
```


Step 2: EER file splitting

Modify the split script **eer_writer_5_v4_py34.py**. Based on the results obtained in **Step 1**, adjust the range parameter in the script to a value marginally exceeding the **Step 1** result.

```
if __name__ == "__main__":
    #Falcon4i
    first_range = (380, 500)
    blank_range = (35, 55)
    end_range = (40, 85)
```

Then run:

```
python3 eer_writer_5_v4_py34.py
```

```
usage: eer_writer_5_v4_py34.py [-h] frame_num
eer_writer_5_v4_py34.py: error: the following arguments are required: frame_num
```

The **Frame_num** parameter should be set to the EER frame number specified during data collection; this will determine the frame count of the split output files.

```
20260109_170347_EER_GainReference.gain
calculate_sep_range_python34.py
eer_name.lst
eer_writer_5_v4_py34.py
failed_Falcon4_00092.eer
failed_Falcon4_00144.eer
failed_Falcon4_00584.eer
Falcon4_00000.eer
Falcon4_00001.eer
Falcon4_00002.eer
Falcon4_00003.eer
Falcon4_00004.eer
Falcon4_00005.eer
Falcon4_00006.eer
Falcon4_00007.eer
Falcon4_00008.eer
Falcon4_00009.eer
Falcon4_00010.eer
Falcon4_00011.eer
Falcon4_00012.eer
Falcon4_00013.eer
Falcon4_00014.eer
Falcon4_00015.eer
Falcon4_00016.eer
Falcon4_00017.eer
Falcon4_00018.eer
Falcon4_00019.eer
Falcon4_00020.eer
Falcon4_00021_00.eer
Falcon4_00021_01.eer
Falcon4_00021_02.eer
Falcon4_00021_03.eer
Falcon4_00021_json
Falcon4_00021_json_dose_sep
Falcon4_00021log_sep.txt
Falcon4_00022_00.eer
Falcon4_00022_01.eer
Falcon4_00022_02.eer
Falcon4_00022_03.eer
Falcon4_00022_json
Falcon4_00022_json_dose_sep
Falcon4_00022log_sep.txt
```

Note on output files:

- BIS output files remain unchanged.
- CR-BIS files are split into individual files for each target position, and the original files are deleted.
- The ***.eer_log** file is replaced with ***log_sep.txt**, which will be used for subsequent .mdoc file generation.

```
cat Falcon4_00021log_sep.txt
```

```
5.99553 Falcon4_00021_00.eer -0.284781 -1.791169 4 0.041321 6 34 R 29471278
5.99553 Falcon4_00021_01.eer -1.637413 1.343705 5 0.020546 6 34 R 29471278
5.99553 Falcon4_00021_02.eer 2.867066 -0.467366 6 0.016017 6 34 R 29471278
5.99553 Falcon4_00021_03.eer 1.537317 2.590912 7 0.007401 6 34 R 29471278
```

CR-BIS-Tomo Output Folder Example (After Splitting)

3.3 Cryo-ET mdoc File Generation

During data acquisition, collection parameters are written into a **tomo_log.txt file**. An example of the original **tomo_log.txt** is shown below:

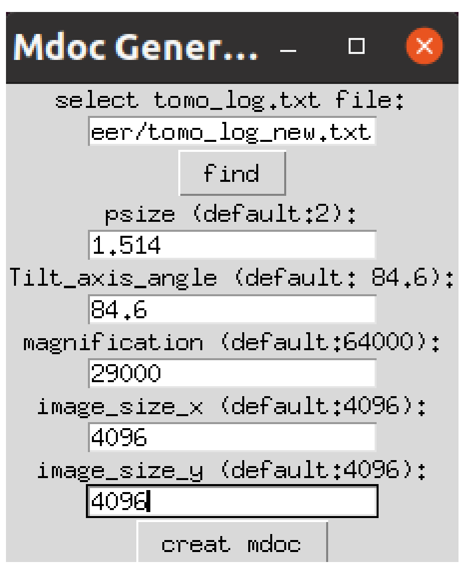
```
(base) yq@IBPserver192:/data42/YQ/Falcon_data/20260113_fast_tomo/raw_eer$ more tomo_log.txt
tomo_start at Nav_Item 3
-0.004235 Falcon4_00001.eer -0.287855 -1.964547 4 0.043952 1 39 R 29471278
-0.004235 Falcon4_00002.eer -1.649436 1.189066 5 0.042708 1 39 R 29471278
-0.004235 Falcon4_00003.eer 2.857132 -0.626617 6 -0.097707 1 39 R 29471278
-0.004235 Falcon4_00004.eer 1.5175 2.452106 7 -0.13598 1 39 R 29471278
1.99552 Falcon4_00005.eer -0.335217 -1.827753 4 -0 2 38 R 29471278
1.99552 Falcon4_00006.eer -1.695808 1.323987 5 -0 2 38 R 29471278
1.99552 Falcon4_00007.eer 2.81185 -0.493756 6 -0 2 38 R 29471278
1.99552 Falcon4_00008.eer 1.473748 2.582073 7 -0 2 38 R 29471278
-2.00399 Falcon4_00009.eer -0.302623 -1.8338 4 0.075467 3 37 R 29471278
-2.00399 Falcon4_00010.eer -1.663199 1.317912 5 0.078269 3 37 R 29471278
-2.00399 Falcon4_00011.eer 2.839884 -0.49118 6 0.059109 3 37 R 29471278
-2.00399 Falcon4_00012.eer 1.500688 2.586718 7 0.063084 3 37 R 29471278
-4.004744 Falcon4_00013.eer -0.290261 -1.829762 4 0.049585 4 36 R 29471278
-4.004744 Falcon4_00014.eer -1.648848 1.3178 5 0.023153 4 36 R 29471278
-4.004744 Falcon4_00015.eer 2.849413 -0.48091 6 0.030499 4 36 R 29471278
-4.004744 Falcon4_00016.eer 1.512214 2.592982 7 0.022223 4 36 R 29471278
3.995775 Falcon4_00017.eer -0.281635 -1.800779 4 0.037872 5 35 R 29471278
3.995775 Falcon4_00018.eer -1.639578 1.345347 5 0.039278 5 35 R 29471278
3.995775 Falcon4_00019.eer 2.867881 -0.470791 6 0.029663 5 35 R 29471278
3.995775 Falcon4_00020.eer 1.533086 2.598611 7 0.031658 5 35 R 29471278
5.99553 Falcon4_00021.eer 1.537317 2.590912 3 7 6 35 R 29471278
-6.004 Falcon4_00022.eer 1.528771 2.585431 3 7 7 34 R 29471278
-8.004254 Falcon4_00023.eer 1.5329 2.587393 3 7 8 33 R 29471278
7.994785 Falcon4_00024.eer 1.562625 2.599155 3 7 9 32 R 29471278
9.99504 Falcon4_00025.eer 1.571911 2.579486 3 7 10 31 R 29471278
-10.004009 Falcon4_00026.eer 1.550648 2.572743 3 7 11 30 R 29471278
-12.004264 Falcon4_00027.eer 1.551012 2.559123 3 7 12 29 R 29471278
11.993795 Falcon4_00028.eer 1.59549 2.571375 3 7 13 28 R 29471278
13.99355 Falcon4_00029.eer 1.592945 2.568637 3 7 14 27 R 29471278
-14.004519 Falcon4_00030.eer 1.577725 2.518067 3 7 15 26 R 29471278
-16.005274 Falcon4_00031.eer 1.57464 2.49873 3 7 16 25 R 29471278
15.995305 Falcon4_00032.eer 1.623733 2.541011 3 7 17 24 R 29471278
17.99506 Falcon4_00033.eer 1.623813 2.52757 3 7 18 23 R 29471278
-18.005029 Falcon4_00034.eer 1.588438 2.476088 3 7 19 22 R 29471278
-20.005284 Falcon4_00035.eer 1.581466 2.467673 3 7 20 21 R 29471278
19.992816 Falcon4_00036.eer 1.64729 2.500581 3 7 21 20 R 29471278
21.99307 Falcon4_00037.eer 1.646323 2.477845 3 7 22 19 R 29471278
```

For EER-format data:

Run the script **tomo_log_edit.py**, which reads the updated information from the corresponding ***log_sep.txt** file (generated after splitting) and modifies **tomo_log.txt** accordingly. The output is saved as **tomo_log_new.txt**, with a format illustrated below:

```
(base) yq@IBPserver192:/data42/YQ/Falcon_data/20260113_fast_tomo/raw_eer$ more tomo_log_new.txt
tomo_start at Nav_Item 3
-0.004235 Falcon4_00001.eer -0.287855 -1.964547 4 0.043952 1 39 R 29471278
-0.004235 Falcon4_00002.eer -1.649436 1.189066 5 0.042708 1 39 R 29471278
-0.004235 Falcon4_00003.eer 2.857132 -0.626617 6 -0.097707 1 39 R 29471278
-0.004235 Falcon4_00004.eer 1.5175 2.452106 7 -0.13598 1 39 R 29471278
1.99552 Falcon4_00005.eer -0.335217 -1.827753 4 -0 2 38 R 29471278
1.99552 Falcon4_00006.eer -1.695808 1.323987 5 -0 2 38 R 29471278
1.99552 Falcon4_00007.eer 2.81185 -0.493756 6 -0 2 38 R 29471278
1.99552 Falcon4_00008.eer 1.473748 2.582073 7 -0 2 38 R 29471278
-2.00399 Falcon4_00009.eer -0.302623 -1.8338 4 0.075467 3 37 R 29471278
-2.00399 Falcon4_00010.eer -1.663199 1.317912 5 0.078269 3 37 R 29471278
-2.00399 Falcon4_00011.eer 2.839884 -0.49118 6 0.059109 3 37 R 29471278
-2.00399 Falcon4_00012.eer 1.500688 2.586718 7 0.063084 3 37 R 29471278
-4.004744 Falcon4_00013.eer -0.290261 -1.829762 4 0.049585 4 36 R 29471278
-4.004744 Falcon4_00014.eer -1.648848 1.3178 5 0.023153 4 36 R 29471278
-4.004744 Falcon4_00015.eer 2.849413 -0.48091 6 0.030499 4 36 R 29471278
-4.004744 Falcon4_00016.eer 1.512214 2.592982 7 0.022223 4 36 R 29471278
3.995775 Falcon4_00017.eer -0.281635 -1.800779 4 0.037872 5 35 R 29471278
3.995775 Falcon4_00018.eer -1.639578 1.345347 5 0.039278 5 35 R 29471278
3.995775 Falcon4_00019.eer 2.867881 -0.470791 6 0.029663 5 35 R 29471278
3.995775 Falcon4_00020.eer 1.533086 2.598611 7 0.031658 5 35 R 29471278
5.99553 Falcon4_00021_00.eer -0.284781 -1.791169 4 0.041321 6 34 R 29471278
5.99553 Falcon4_00021_01.eer -1.637413 1.343705 5 0.020546 6 34 R 29471278
5.99553 Falcon4_00021_02.eer 2.867066 -0.467366 6 0.016017 6 34 R 29471278
5.99553 Falcon4_00021_03.eer 1.537317 2.590912 7 0.007401 6 34 R 29471278
-6.004 Falcon4_00022_00.eer -0.273592 -1.834182 4 0.001281 7 33 R 29471278
-6.004 Falcon4_00022_01.eer -1.628307 1.304378 5 0.002839 7 33 R 29471278
-6.004 Falcon4_00022_02.eer 2.86223 -0.480364 6 0.004551 7 33 R 29471278
-6.004 Falcon4_00022_03.eer 1.528771 2.585431 7 0.005535 7 33 R 29471278
-8.004254 Falcon4_00023_00.eer -0.272381 -1.826702 4 0 8 32 R 29471278
-8.004254 Falcon4_00023_01.eer -1.620628 1.299627 5 0 8 32 R 29471278
-8.004254 Falcon4_00023_02.eer 2.860319 -0.466981 6 0 8 32 R 29471278
-8.004254 Falcon4_00023_03.eer 1.5329 2.587393 7 0 8 32 R 29471278
7.994785 Falcon4_00024_00.eer -0.268956 -1.764994 4 -0 9 31 R 29471278
7.994785 Falcon4_00024_01.eer -1.614469 1.356418 5 -0 9 31 R 29471278
7.994785 Falcon4_00024_02.eer 2.885041 -0.445256 6 -0 9 31 R 29471278
7.994785 Falcon4_00024_03.eer 1.562625 2.599155 7 -0 9 31 R 29471278
```

In this new file, the CR-BIS-related file information is replaced with the paths and details of the split output files. Subsequently, run **mdoc.py** to generate the **.mdoc** file for each tilt series.



For TIFF-format data:

First, execute **modi_tif_log.py** to create an updated ***modi.tif_log** file. Then, run **tomo_log_edit.py** to modify **tomo_log.txt** using the updated log. Finally, use **mdoc.py** to generate the **.mdoc** files for each tilt series.